THE PHANTOM BARRELS

How stockpiling inflated China's apparent oil demand, 2022 to July 2026, and what the Hormuz shutdown revealed

China cut crude imports from 11.7 mb/d in February 2026 to 7.8 mb/d in May, a decade low, with no domestic fuel crisis [1,2]. That outcome is only consistent with one reading of the prior three years: a large share of China's headline import demand was inventory accumulation, not consumption. Real transport-fuel demand peaked in 2023 and declined thereafter [7,8,10]. Implied stock builds of roughly 0.75 to 1.1 mb/d in 2025 accounted for effectively all marginal import growth [4,5]. Total crude stocks reached an estimated 1.2 to 1.4 billion barrels by end-2025 [4,5,6]. The buildup accelerated in step with Project 2025's publication, the 2024 US election, and China's Energy Law, and it is the primary reason Brent held near $100 through the largest supply disruption in oil-market history [1,2,3]. By early July, a US-Iran memorandum with a temporary sanctions waiver reopened the strait faster than expected: Brent returned to pre-war levels near $72 [20,21], while June seaborne imports fell further to 6.4 mb/d, the lowest since 2016 [23], as Beijing lets the glut build ahead of an expected August restocking pulse [22].

1. The wedge

Figure 1 reconstructs the balance. Apparent crude demand is imports plus domestic output. Estimated true consumption subtracts implied stock change. The gap between the curves is the stockpile build. The wedge widens from 2024, peaks in late 2025 (the November 1.88 mb/d storage surplus [11]), and inverts after February 28, 2026, when the war begins and China starts running on stored barrels.

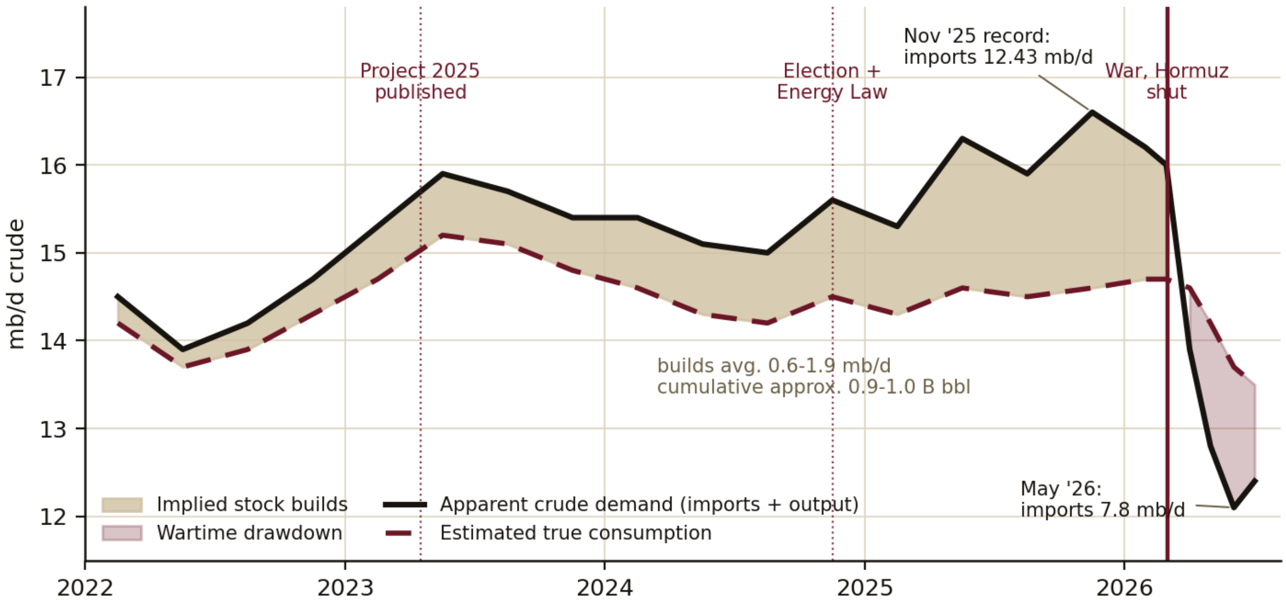


Figure 1. China apparent crude demand vs. estimated true consumption, quarterly reconstruction. Anchors: GAC customs and NBS via [1,2,11]; build estimates [4,5]. Series are reconstructions, not official data.

2. Real consumption peaked before imports did

Combined gasoline, diesel and jet demand fell in 2024 to about 8.1 mb/d, 2.5% below 2021 and barely above 2019 [7]. Apparent diesel demand dropped from 4.7 mb/d in April 2023 to 4.0 mb/d in April 2025 [8]. Gasoline demand fell 14% year over year in August 2024 [10]. Refinery runs fell 3.3% in 2024 [12]. Sinopec and CNPC research institutes both date the transport-fuel peak to 2023 [13]. Petrochemical feedstock was the only growing segment, up about 5% in 2024 [7].

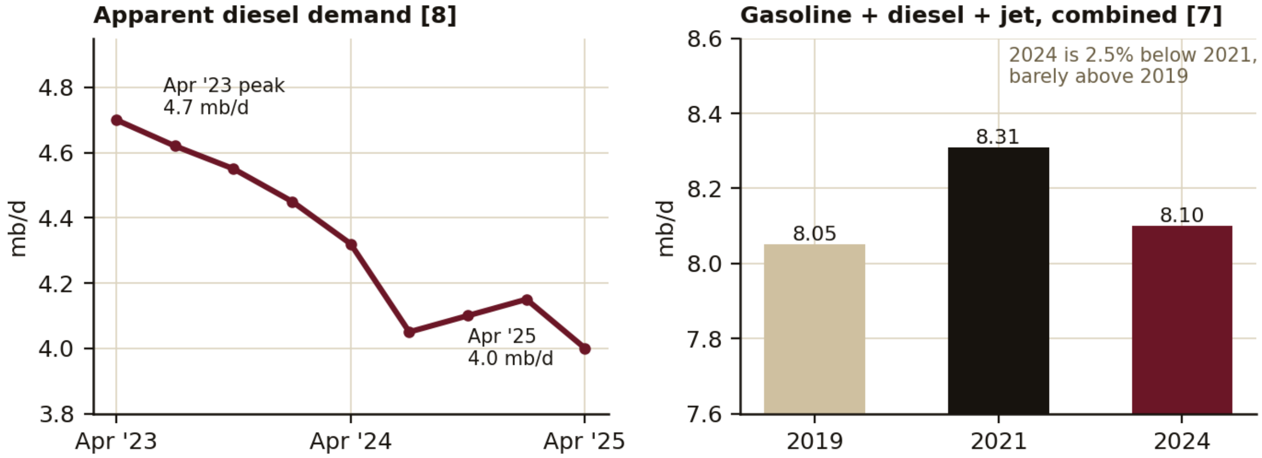


Figure 2. Left: apparent diesel demand [8]. Right: combined transport fuels by year [7].

The decline is structural, not cyclical. NEVs reached roughly half of new car sales by 2025; LNG trucks went from 9% of heavy-duty sales in 2022 to 42% by 2024; high-speed rail and metro expansion displaced road and air fuel [7,9]. The IEA estimates substitution avoided about 1.2 mb/d of demand between 2019 and 2024 [7]. GDP growth of about 4 to 4.7% [10] ran well below the pre-2020 trend, and the property slump removed the largest single driver of diesel use [9].

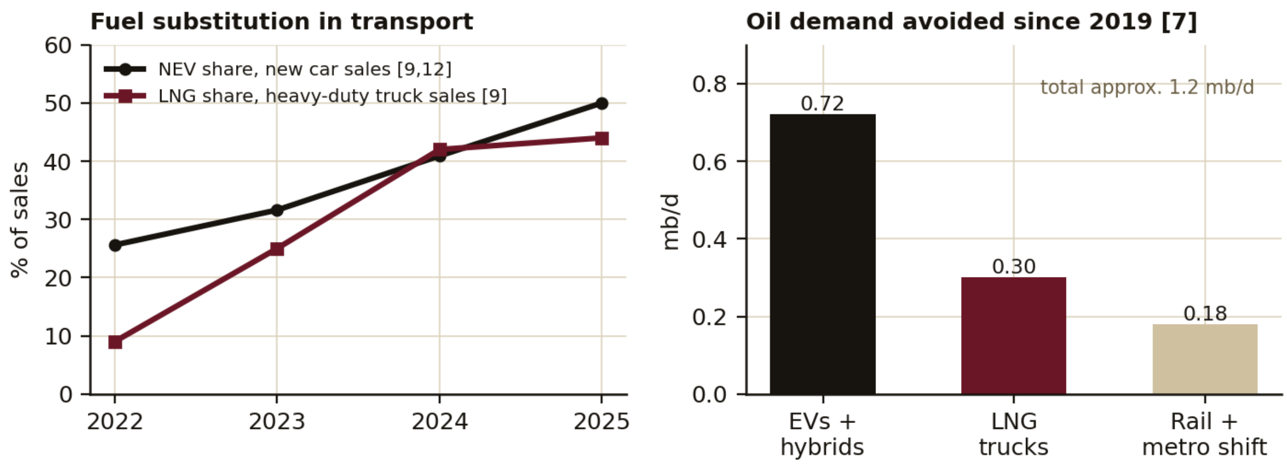


Figure 3. Left: substitution shares in vehicle sales [9,12]. Right: decomposition of avoided oil demand since 2019 [7]. Decomposition split is approximate.

3. The builds, quantified

EIA estimates China added an average of 1.1 mb/d to inventories in 2025; OIES puts it at 0.75 mb/d, within a published range of 0.4 to 1.1 [4,5]. Government-held stocks were about 360 million barrels in December 2025, with commercial stocks near 1.0 billion, and national oil companies have been directed since 2024 to hold emergency oil in commercial tanks, making the commercial layer a de facto second SPR [4]. The Energy Law, passed in November 2024 and effective January 2025, put corporate stockpiling into statute [5]. Against real consumption that was flat to falling, builds of this size were effectively all of China's marginal crude demand in 2024 and 2025, roughly 7 to 10% of headline imports.

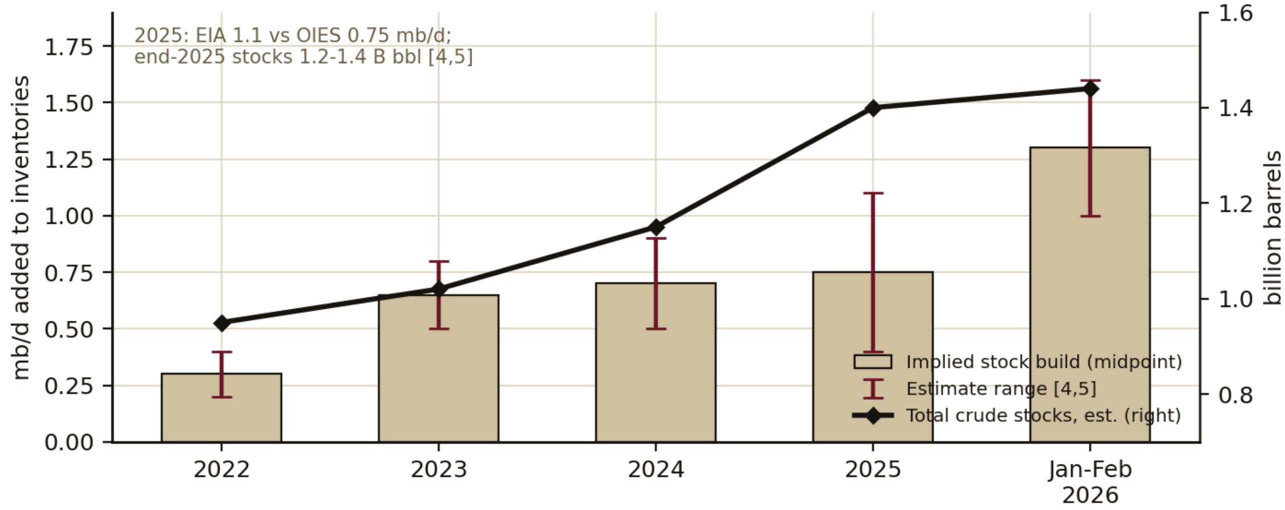


Figure 4. Implied annual stock builds with estimate ranges (bars, left axis) and estimated total crude stocks (line, right axis) [4,5,6]. Jan-Feb 2026 is a two-month annualized rate.

4. Timing against Project 2025

Mandate for Leadership was published in April 2023. Its energy chapter frames American oil exports as foreign-policy leverage that will, in its words, "provide tools for U.S. policymakers to assist our allies and deter our adversaries" [14]. The Western Hemisphere chapter argues for re-hemisphering energy supply away from distant suppliers it calls manipulable, and one passage names the five countries deserving the next administration's attention and energy: China, Iran, Russia, Venezuela and North Korea [14,15]. Three of those five were China's discounted crude suppliers, providing roughly 48% of Shandong independent refiner imports in 2025 [5]. A June 2024 Baker Institute testimony to Congress identified energy stockpiling as the classic strategic warning indicator of a state preparing for conflict [6]. The sequence of Chinese policy actions and the buildup itself track the US political timeline:

Date	Event	Reading
Apr 2023	Project 2025 published; five priority countries named [14,15]	**The telegraph**
Apr 2023	Diesel demand peaks at 4.7 mb/d, structural decline begins [8]	**Real demand rolls over**
2023	China absorbs 1+ mb/d of discounted, mostly Russian crude [16]	**Opportunistic phase**
2024	NOCs directed to hold emergency oil in commercial tanks [4]	**Shadow SPR stood up**
Nov 2024	Trump elected; Energy Law passed the same month [5]	**Stockpiling codified**
Jan 2025	Energy Law effective; US energy-dominance orders signed	**Both sides commit**
Jun 2025	12-day Israel-Iran air war; refiners buy nothing into the spike [17]	**Reserve already deep**
Nov 2025	Imports hit a 27-month record 12.43 mb/d; 1.88 mb/d to tanks [11]	**Final acceleration**
Jan 3, 2026	Maduro captured; Venezuelan flow rerouted to authorized channels	**First supplier cut**
Feb 15-20, 2026	Iran triples export rate days before the strikes [3]	**Frontloading**
Feb 28, 2026	War begins; Iran shuts Hormuz Mar 4; ~10 mb/d of exports lost [3]	**The tail event**
May 2026	Imports at 7.8 mb/d, a decade low; 74% of the global decline [1,2]	**Reserve does its job**

Table 1. Event sequence, April 2023 to May 2026.

5. The 2026 test

The war removed about 10 mb/d of Persian Gulf exports, the largest disruption on record [3]. China's response was to stop buying rather than bid for scarce barrels: imports fell 3.9 mb/d from February to May, which JPMorgan estimates was 74% of the entire global decline in crude trade [1]. Domestic Chinese product prices moved at roughly one fifth of Brent's volatility through the March spike [18]. Morgan Stanley called the import cut the single most important reason prices are not higher [2]. A demand cut of that size, absorbed without rationing, is direct evidence that the marginal import barrels of 2024-2025 were never consumption.

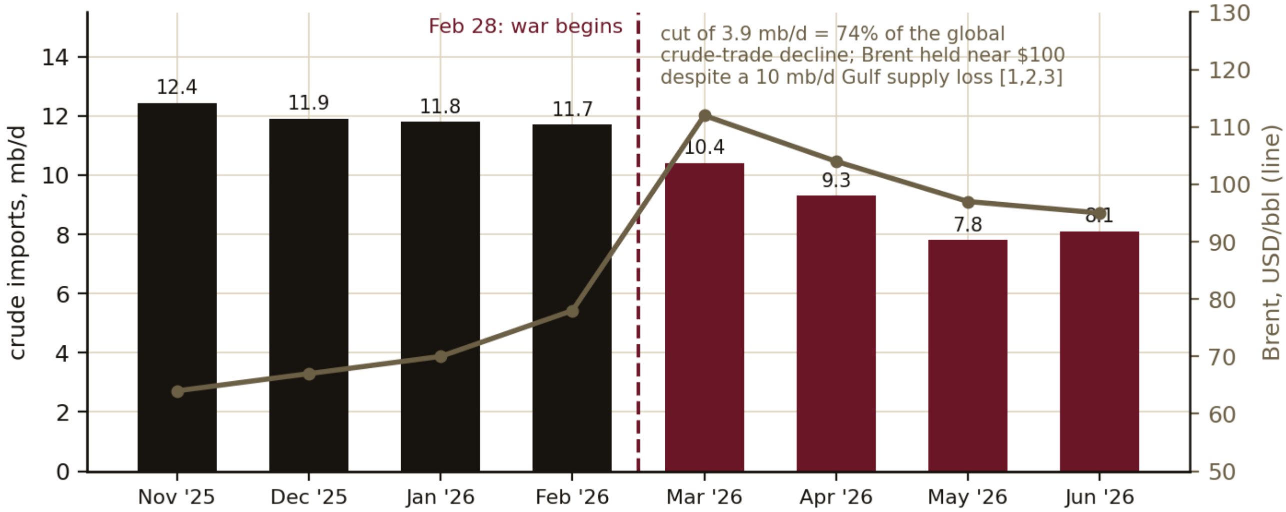


Figure 5. Monthly crude imports (bars) against Brent (line), November 2025 to June 2026. Nov, Feb, May and June values are reported [1,2,11,23]; June combines seaborne arrivals with estimated overland flows; other months are interpolated from reported trajectories.

6. Assessment

The stockpiling-inflated-demand mechanism is real and quantifiable: a wedge of roughly 1 mb/d that was effectively 100% of marginal import growth in 2024-2025, cumulating to an estimated 0.9 to 1.0 billion barrels since 2022 against total-stock estimates of 1.2 to 1.4 billion. The timing correlation with Project 2025, the election and the Energy Law is strong. The defensible attribution is general war-proofing against the world Project 2025 described (sanctions risk, supplier decapitation, Taiwan optionality) rather than specific foreknowledge of the Venezuela or Hormuz operations. Beijing hedged the distribution of outcomes; the tail event happened.

Two caveats bound the bullish read. First, the 1.4 billion barrels nominally cover about 120 days of imports but only 60 to 90 days of working stock before refiners cut runs to protect tank minimums [3]. Second, the import cut was not purely voluntary: the blockade and the US counter-blockade physically removed the discounted barrels. Both imply eventual import snap-back plus mandatory restocking demand once the strait normalizes, a structural support for crude into 2027. As of early July that support is a floor under a glutted, reopening market near $72, not a flat-price rally.

7. The route forward

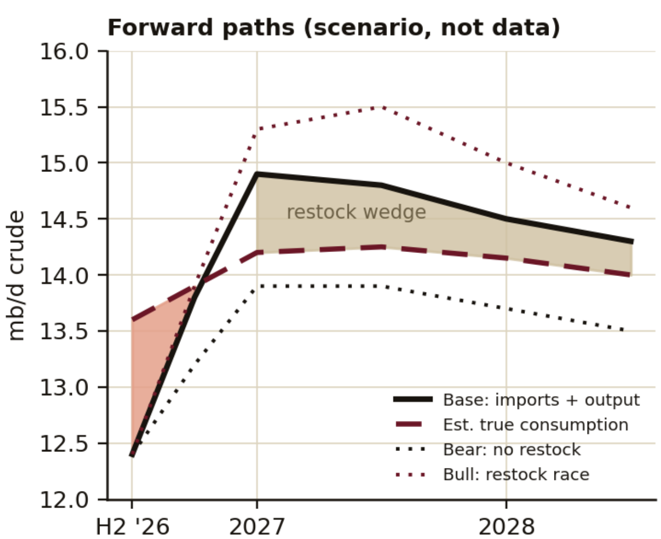
The naive read says less Chinese buying means lower Brent. That holds in one of three configurations, because Chinese imports are endogenous: the level means nothing until conditioned on why. Configuration one, the current tape: buying less because the barrels do not exist. The blockade causes both the low imports and the high price; China's elasticity was the price cap and the main reason Brent peaked at $118 rather than $200 and held near $95 into June [2]. The tape exited configuration one in late June: a US-Iran memorandum, a 60-day framework and a temporary sanctions waiver reopened the strait, and Brent returned to pre-war levels near $72 by July 3 [20,21]. Configuration two, the trapdoor: buying less by choice after normalization. There the intuition is right and prices fall, because consensus demand models still carry roughly 1 mb/d of "demand" that was never consumption. Configuration three: the restock bid, discretionary and price-sensitive, appearing under roughly $80 to 85 and vanishing on rips. It is a floor and a volatility compressor, not a driver. The 2023-2025 phantom era was configuration three running continuously.

The regime update as of July 3: Hormuz flows are back above 10 mb/d with Saudi exports near 90% of baseline and the UAE fully restored; a 54-supertanker backlog is arriving in waves alongside 58 to 68 million barrels of Iranian crude on the water [21]. Morgan Stanley cut forecasts twice in two weeks warning of glut [21]. China moved the other way: June seaborne arrivals of 6.4 mb/d were the lowest since October 2016 [23], consistent with a buyer of China's scale deliberately depressing the price at which it will restock. JPMorgan estimates about 3 mb/d of the import decline is temporary, reversing from August as the chemicals sector returns and state restocking begins, and notes the refined-product export ban still partly in place: fully lifted, product exports could jump 88 to 160% from first-half levels [22]. Spot now sits inside the sub-$70s fill zone that OIES documents as China's historical restock trigger [5].

Quantity	mb/d
True crude consumption, pre-war (refinery runs + direct burn)	**14.5-14.7**
of which crude processed for product exports	**0.9-1.1**
domestic-use crude consumption	**13.5-13.7**
Total liquids demand, CNPC peak (2025) [12]	**15.4**
Maintenance imports (consumption minus 4.3 domestic output, zero stock change)	**10.2-10.4**
Actual imports, 2023-2025 [9,11]	**11.1-11.5**
Standing overshoot = the phantom	**0.8-1.2**

Table 2. Demand arithmetic on a crude basis. Wartime check: run cuts (-0.6), product export bans (-0.3) and substitution (-0.2) put consumption at 13.5-13.8, import requirement 9.2-9.5, actual buying 7.8-8.1, implied draw 1.1-1.7 mb/d, consistent with Figure 1. Cumulative draw through July is roughly 170-190M bbl; a grind into Q1 2027 reaches 400-500M, where the 60-90 day working-stock floor binds [3].

True consumption near 14.5 mb/d implies the market never needed China above about 10.3 mb/d of imports; everything above that was policy demand. The single most informative datapoint of H2 2026 will be which run-rate customs prints in the first two clean months after the strait functions: about 10.2 means no restock and consensus demand overstated by 1.0 to 1.3 mb/d; 10.6 a 24-month rebuild of roughly 250M bbl; 11.0 a 12-month rebuild; a sustained 11.0 to 11.6 means a buffer-expansion doctrine that recreates the phantom optics for a second time. Storage headroom exists: above-ground tanks were only 56% filled as of March 2025 [9].



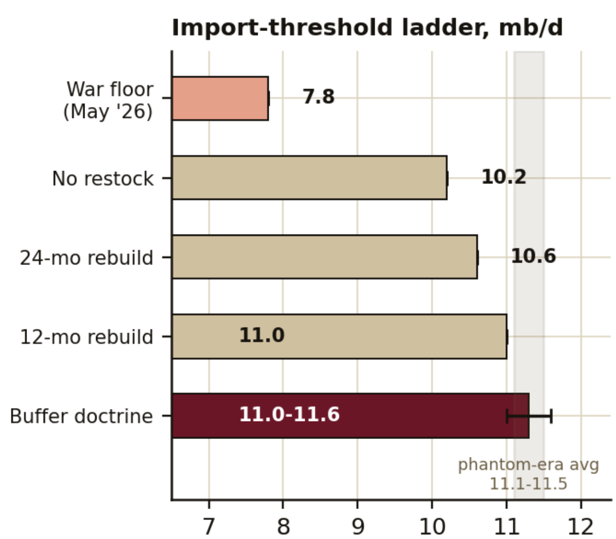


Figure 6. Left: forward paths for apparent demand against estimated true consumption; the shaded area is the projected restock wedge in the base case. Right: post-normalization import run-rates and what each implies. All values are scenario projections, not data.

Scenarios, reweighted for the July tape. Bear, roughly 30-35%, partially realized: the phantom unwind. Full reopening, a permanent waiver, no urgent restock, de-oiling as doctrine, OPEC+ spare and Venezuelan supply returning. The market reprices to true consumption near 14.5 and falling with the standing 1 mb/d stockpile bid gone; the balance flips to a 1.5 to 2.5 mb/d surplus in 2027. Brent 52-65 in contango; the first leg already printed with Brent down 25% from the June highs. The product-export ban is the crack trigger: fully lifted, Singapore gasoil compresses to $5-10 as Chinese exports surge; Atlantic diesel holds $15-20 on European closures; jet is the relative winner. Base, roughly 45%: the August restock pulse lands as JPMorgan expects, the tanker backlog digests in waves, and a war-risk premium lingers while the 60-day framework holds. China restocks 0.4-0.8 mb/d at the fill zone (imports recovering toward 10.5-11.2) while the US, Japan, Korea and India rebuild against historically thin cover [2,19]. Brent 62-80 with a hard floor at the fill zone and a soft ceiling. Cracks outlast crude with the export ban partially in place: diesel $14-22, jet $16-24, gasoline $10-15, normalizing through mid-2027. Bull, roughly 20-25%: the restock race, on either of two triggers. The framework fails (Iran is demanding strait tolls and claiming control of the lane; full 20 mb/d throughput is unlikely before 2027 given damage [21]) and re-escalation reprices Brent to 90-120. Or it holds and coordinated Chinese, US, Japanese, Korean and Indian restocking of 1.5-2.5 mb/d for a year or more meets the market at 85-100. Cracks blow out first (diesel $35-50), then roll over before flat price; cracks weakening while Brent holds is the exit signal.

True consumption trends down in every branch: transport fuels decline 1 to 2% a year on substitution, and petrochemical growth sits on light feedstocks rather than crude-intensive runs [7,12]. What differs across scenarios is the import path around consumption: the timing and size of the restock wedge. The tells are the August and September customs prints against the 10.2 / 10.6 / 11.0 / 11.5 thresholds, and the product-export ban, which reprices cracks before crude if fully lifted. Positioning follows from that wedge, not from a China growth story.

8. Method notes

China publishes no oil inventory data. Curves in Figure 1 are quarterly reconstructions: apparent demand is imports plus domestic production; true consumption is apparent demand minus implied stock change, where the stock change is bounded by published estimates (2025 range 0.4 to 1.1 mb/d; OIES 0.75, EIA 1.1). The wedge shown is toward the upper bound of that range, and monthly 2026 values other than reported anchors are interpolated. Figures are illustrative of cited anchors, not official statistics. Where estimates conflict, both are shown or the range is stated. The route-forward section is scenario projection with subjective probabilities, not data.

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